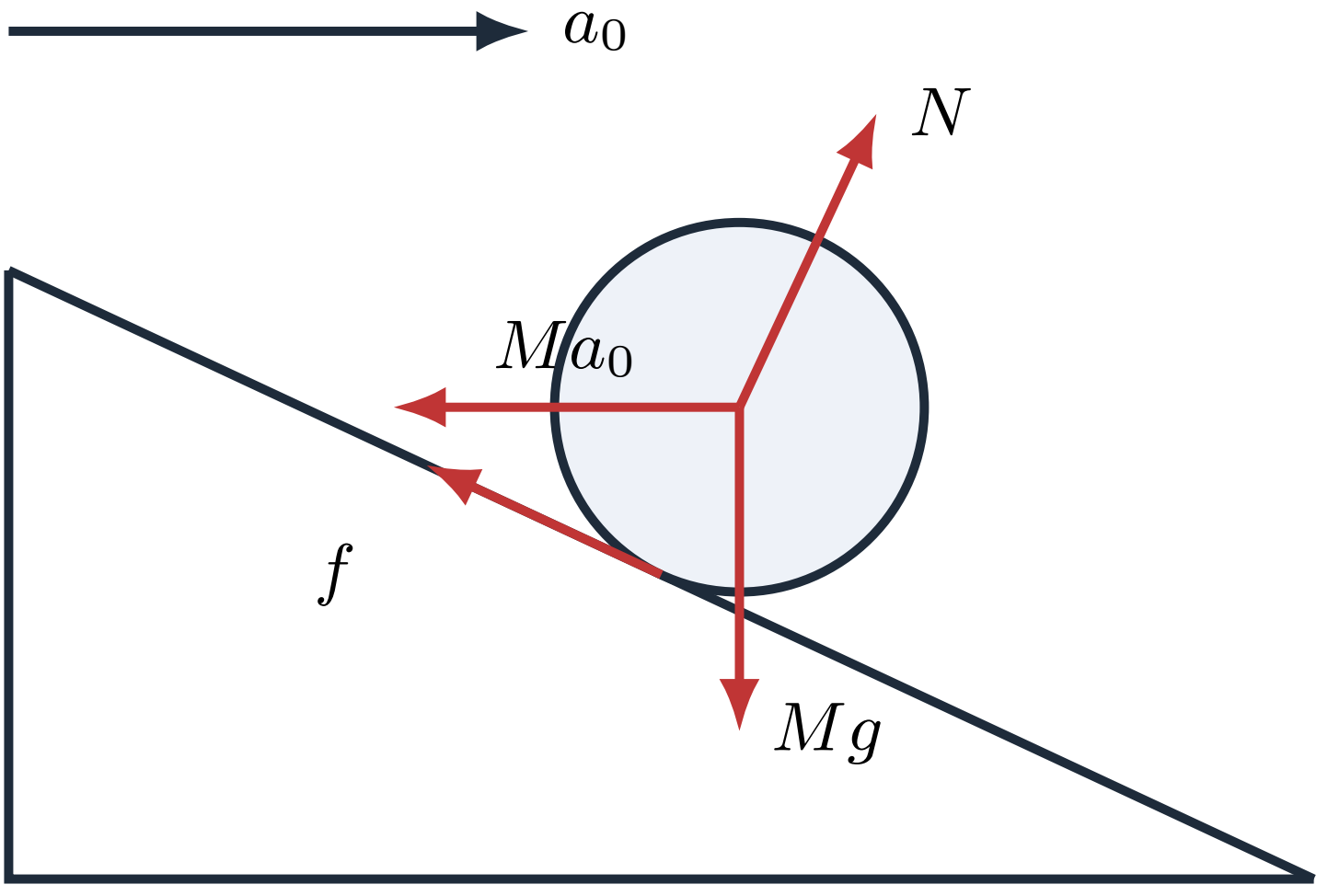

$$a_r = \tfrac{2}{3}(g\sin\theta - a_0\cos\theta)$$